

MORELIA, Mexico

WMO: 766650

Lat: 19.7217N

Lon: 101.1831W

Elev: 6276

StdP: 11.65

Time Zone: -6.00 (MEX)

Period: 83-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	43.3	45.2	29.1	29.2	68.8	32.2	33.7	65.4	18.7	64.5	15.8	71.0	2.1	210	0.302

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	26.0	88.9	61.9	86.6	62.1	84.4	61.6	68.5	78.9	67.3	77.9	66.1	77.0	4.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	118.9	74.7	63.9	112.6	72.9	62.7	107.8	71.4	36.8	78.3	35.6	78.3	34.6	76.8	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	13.5	10.8	8.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)				N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.0	59.9	62.9	65.6	70.2	71.5	69.9	67.1	67.9	67.5	65.9	62.5	60.9
	DBStd	4.75	3.01	3.42	4.06	3.22	3.26	3.43	2.01	1.98	2.06	3.08	3.54	3.25
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	534	158	73	39	3	1	2	6	3	5	27	87	130
	CDD50	5843	307	361	483	607	666	598	531	556	526	494	376	338
	CDD65	900	0	14	56	160	202	150	72	94	81	55	13	3
	CDH74	8026	105	384	854	1907	1947	1026	352	459	314	361	197	120
	CDH80	1848	1	38	159	677	633	264	16	17	8	22	9	4
Wind	WSAvg	3.3	3.6	3.9	3.8	3.7	3.2	3.4	3.0	2.9	3.1	3.3	3.2	3.1
Precipitation	PrecAvg	38.2	0.7	0.6	0.6	0.4	2.0	6.4	8.5	7.8	6.9	3.0	0.7	0.6
	PrecMax	48.8	6.5	7.2	6.7	1.6	5.2	10.0	16.4	12.9	10.6	5.9	2.4	8.7
	PrecMin	25.9	0.0	0.0	0.0	0.0	0.0	2.4	3.2	4.7	3.7	0.8	0.1	0.0
	PrecStd	6.0	1.3	1.4	1.3	0.5	1.4	1.9	2.6	2.1	1.8	1.4	0.7	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.3	83.4	87.4	91.4	92.2	88.9	82.0	82.0	81.2	82.5	81.3	80.1
		MCWB	58.4	59.8	62.4	62.1	62.9	64.3	64.6	64.9	65.9	63.8	61.9	60.8
	2%	DB	76.7	80.9	84.1	88.9	89.1	85.6	79.6	80.0	79.1	79.9	78.6	77.1
		MCWB	57.6	59.1	61.6	61.1	61.8	64.1	64.8	64.9	64.7	63.0	61.2	60.1
	5%	DB	74.5	78.5	81.6	86.6	86.5	83.1	77.8	78.6	77.4	78.0	76.3	74.7
		MCWB	57.3	58.4	60.5	60.9	62.3	64.0	64.6	64.6	64.2	62.3	60.5	58.9
	10%	DB	72.2	76.0	79.2	84.2	84.0	80.3	75.7	77.0	75.8	75.9	73.9	72.3
		MCWB	56.3	57.2	59.2	60.6	61.1	63.9	64.2	64.5	63.9	61.7	59.8	57.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.6	63.5	67.3	66.6	68.4	70.9	69.1	69.4	69.2	67.9	66.0	64.2
		MCDB	73.7	79.2	81.7	85.8	83.5	81.7	77.1	78.0	76.8	77.7	75.9	73.9
	2%	WB	59.4	61.2	64.7	64.5	66.4	67.9	67.2	67.6	67.3	66.0	64.1	62.3
		MCDB	73.5	77.0	80.0	84.3	81.1	79.7	76.1	77.0	76.1	75.9	73.9	73.0
	5%	WB	58.2	59.5	62.6	63.1	65.1	66.1	65.7	66.0	65.9	64.6	62.6	60.8
		MCDB	72.3	75.5	78.5	82.8	80.0	77.3	75.1	75.8	74.9	74.6	72.5	72.0
	10%	WB	56.9	58.1	60.5	61.6	63.8	64.8	64.5	64.7	64.7	63.4	61.1	59.1
		MCDB	70.4	73.5	76.7	80.6	78.7	76.0	74.2	74.7	73.5	72.7	71.4	70.6
Mean Daily Temperature Range	5% DB	MDBR	26.9	27.9	28.5	30.4	26.0	20.1	18.8	19.0	18.4	22.2	25.3	26.1
		MCDBR	30.4	31.6	31.9	33.4	29.8	24.5	21.8	21.4	20.9	25.1	28.0	29.2
		MCWBR	13.9	13.1	13.7	12.7	10.8	8.6	9.6	8.8	8.7	10.5	12.7	14.1
	5% WB	MCDBR	28.4	29.4	29.9	30.9	25.4	21.4	19.8	20.3	19.4	22.2	25.1	27.7
		MCWBR	13.9	13.0	14.8	13.1	10.5	9.5	10.4	10.0	9.6	10.4	12.9	14.7
Clear-Sky Solar Irradiance	taub	0.331	0.340	0.334	0.377	0.460	0.419	0.404	0.401	0.404	0.373	0.331	0.328	
	taud	2.427	2.393	2.406	2.302	2.096	2.247	2.300	2.306	2.303	2.389	2.463	2.462	
	Ebn at Noon	301	304	309	294	267	276	280	283	283	289	299	298	
	Edh at Noon	34	37	38	43	52	44	42	42	42	37	32	32	
All-Sky Solar Radiation	RadAvg	1538	1827	2115	2266	2087	1826	1781	1783	1612	1633	1588	1503	
	RadStd	115	167	119	82	128	121	113	98	102	108	120	84	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	+1.44	-3.11	N/A	-1.33	-1.69	-54	N/A	N/A	N/A

Nomenclature: See separate page